An Analysis of Mixture of experts

Mixture of experts

Mixture of experts -- Part 1

Experts f 1 , . . . , f n {\displaystyle f_{1},...,f_{n}} , each taking the same input x {\displaystyle x} , and producing outputs f 1 (x) , . . . , f n (x) {\displaystyle f_{1}(x),...,f_{n}(x)} . A weighting function (also known as a gating function) w {\displaystyle w} , which takes input x {\displaystyle x} and produces a vector of outputs (w (x) 1 , . . . , w (x) n) {\displaystyle (w(x)_{1},...,w(x)_{n})} . This may or may not be a probability distribution, but in both cases, its entries are non-negative.

Mixture of experts -- Part 2

Meta-pi network The meta-pi network, reported by Hampshire and Waibel, uses f (x) = ∑ i w (x) i f i (x) {\displaystyle f(x)=\sum _{i}w(x)_{i}f_{i}(x)} as the output. The model is trained by performing gradient descent on the mean-squared error loss L := 1 N ∑ k ■ y k − f (x k) ■ 2 {\displaystyle L:={\frac {1}{N}}\sum _{k}\|y_{k}-f(x_{k})\|^{2}} . The experts may be arbitrary functions. In their original publication, they were solving the problem of classifying phonemes in speech signal from 6 different Japanese speakers, 2 females and 4 males. They trained 6 experts, each being a "time-delayed neural network" (essentially a multilayered convolution network over the mel spectrogram). They found that the resulting mixture of experts dedicated 5 experts for 5 of the speakers, but the 6th (male) speaker does not have a dedicated expert, instead his voice was classified by a linear combination of the experts for the other 3 male speakers.

Mixture of experts -- Part 3

Deep learning The previous section described MoE as it was used before the era of deep learning. After deep learning, MoE found applications in running the largest models, as a simple way to perform conditional computation: only parts of the model are used, the parts chosen according to what the input is. The earliest paper that applies MoE to deep learning dates back to 2013, which proposed to use a different gating network at each layer in a deep neural network. Specifically, each gating is a linear-ReLU-linear-softmax network, and each expert is a linear-ReLU network. Since the output from the gating is not sparse, all expert outputs are needed, and no conditional computation is performed. The key goal when using MoE in deep learning is to reduce computing cost. Consequently, for each query, only a small subset of the experts should be queried. This makes MoE in deep learning different from classical MoE. In classical MoE, the output for each query is a weighted sum of all experts' outputs. In deep learning MoE, the output for each query can only involve a few experts' outputs. Consequently, the key design choice in MoE becomes routing: given a batch of queries, how to route the queries to the best experts.

Mixture of experts -- Part 4

Adaptive mixtures of local experts The adaptive mixtures of local experts uses a Gaussian mixture model. Each expert simply predicts a Gaussian distribution, and totally ignores the input. Specifically, the i {\displaystyle i} -th expert predicts that the output is y ∼ N (μ i , I) {\displaystyle y\sim N(\mu _{i},I)} , where μ i {\displaystyle \mu _{i}} is a learnable parameter. The weighting function is a linear-softmax function: w (x) i = e k i T x + b i ∑ j e k j T x + b j {\displaystyle w(x)_{i}={\frac {e^{k_{i}^{T}x+b_{i}}}{\sum _{j}e^{k_{j}^{T}x+b_{j}}}}} The mixture of experts predict that the output is distributed according to the log-probability density function: ln ■ f θ (y | x) = ln ■ [∑ i e k i T x + b i ∑ j e k j T x + b j N (y | μ i , I)] = ln ■ [(2 π) − d / 2 ∑ i e k i T x + b i ∑ j e k j T x + b j e − 1 2 ■ y − μ i ■ 2] {\displaystyle \ln f_{\theta }(y|x)=\ln \left[\sum _{i}{\frac {e^{k_{i}^{T}x+b_{i}}}{\sum _{j}e^{k_{j}^{T}x+b_{j}}}}N(y|\mu _{i},I)\right]=\ln \left[(2\pi)^{-d/2}\sum _{i}{\frac {e^{k_{i}^{T}x+b_{i}}}{\sum _{j}e^{k_{j}^{T}x+b_{j}}}}e^{-{\frac {1}{2}}\|y-\mu _{i}\|^{2}}\right]} It is trained by maximal likelihood estimation, that is, gradient ascent on f (y | x) {\displaystyle f(y|x)} . The gradient for the i {\displaystyle i} -th expert is

Mixture of experts -- Part 5

Routing In the original sparsely-gated MoE, only the top-k experts are queried, and their outputs are weighted-summed. There are other methods. Generally speaking, routing is an assignment problem: How to assign tokens to experts, such that a variety of constraints are followed (such as throughput, load balancing, etc.)? There are typically three classes of routing algorithm: the experts choose the tokens ("expert choice"), the tokens choose the experts (the original sparsely-gated MoE), and a global assigner matching experts and tokens. During inference, the MoE works over a large batch of tokens at any time. If the tokens were to choose the experts, then some experts might get few tokens, while a few experts get so many tokens that it exceeds their maximum batch size, so they would have to ignore some of the tokens. Similarly, if the experts were to choose the tokens, then some tokens might not be picked by any expert. This is the "token drop" problem. Dropping a token is not necessarily a serious problem, since in Transformers, due to residual connections, if a token is "dropped", it does not disappear. Instead, its vector representation simply passes through the feedforward layer without change. Other approaches include solving it as a constrained linear programming problem, using reinforcement learning to train the routing algorithm (since picking an expert is a discrete action, like in RL). The token-expert match may involve no learning ("static routing"): It can be done by a deterministic hash function or a random number generator.

Mixture of experts -- Part 6

Mixture of experts (MoE) is a machine learning technique where multiple expert networks (learners) are used to divide a problem space into homogeneous regions. MoE represents a form of ensemble learning. They were also called committee machines.